



POLITECNICO MILANO 1863

Requirements Analysis Specification Document

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Students&Companies - Requirements Analysis Specification
Document

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1 Introduction

The Requirement Analysis and Specification Document (RASD) aims to focus on the tasks needed to develop and implement an application, taking account of the requirements of the involved stakeholders and, analyzing and documenting also the application requirements. Then, in the second part of the document, there is a more formal definition of the requirements with the use of the Alloy language. In general this document is meant for developers tasked with the implementation of the system and also for all the other entities involved in validation and managing of the project.

1.1 Purpose

Recently job market becoming more and more competitive. Accordingly students face significant challenges in finding suitable job opportunities and internships with respect to their experiences, skills and most importantly their interest and their career goals. Simultaneously, companies find it hard who with fit their need and requirements as well as their work culture. The Students&Companies (S&C) platform addresses this need by streamlining and enhancing the process of matching students with relevant internship opportunities. By analyzing the experiences, skills and attitudes of students listed in their CVs and comparing them with the specific requirements and perks of available roles provided by the companies, S&C connects students with positions that suit them well.

The platform helps students easily find internship opportunities by collecting detailed information about available internships, student resumes, skills, and preferences. This allows students to search for companies and roles that match their interests through S&C's search system, giving them more control over finding an internship. Additionally, S&C assists companies by managing interview schedules and assessments for students who apply for specific roles, making the hiring process smoother and helping both sides make better decisions. Finally, S&C offers tools to track internships by gathering student feedback and complaints, providing insights to improve future programs. Overall, S&C aims to connect students and companies, making it easier and more transparent for students to start their careers.

This document focuses on the Requirements Analysis and Specification Document (RASD) of the system and describes the main goals, the domain assumptions, the scenarios which may happen, the uses cases, the list of functional and non-functional requirements which system should fulfill, and finally the diagrams to visualize the interactions between components and performance of the system.

1.1.1 Goals

Table 1: Goals Definition

Goals	Description
G1	S&C allows students to manage (create, upload, edit) their resume
G2	S&C allows students to see available internships (e.g., by location, industry)
G3	S&C allows students to initiate the search process and submit applications directly
G4	S&C allows students to track the status (reviewed, shortlisted, selected) of their internship applications
G5	S&C allows students to raise complaints or concerns regarding an internship experience
G6	S&C allows companies to manage (create, edit, publish, delete) internship listings
G7	S&C allows companies to browse and search through student profiles (CVs) applied for the internship (e.g., by location, industry)
G8	S&C allows companies to review incoming applications
G9	S&C allows companies the ability to shortlist candidates
G10	S&C allows companies communicate with applicants
G11	S&C allows companies to provide updates on their application status of incoming applications
G12	S&C allows companies to manage (create, edit, send) questionnaire to applicants
G13	S&C allows companies to manage (scheduling and conducting) interviews with applicants
G14	S&C allows universities to track application processes
G15	S&C allows universities to monitor internship execution
G16	S&C allows universities to collect complaints

1.2 Scope

The Students & Companies (S&C) platform connects and facilitate matching university students want to find internships with companies providing those opportunities. The platform will provide abilities for students to manage their profiles, apply for internships, and track their application status. At the same time, it will allow companies to post internships offerings, review applications, and communicate with potential candidates. The platform also includes an access for universities to monitor internship activities and collect compliance.

To initiate application process, an application has been defined through which the user can do the following:

Student Features

- Students must register to system with their credentials.

- Student must log in to his/her account. Then, students can create, upload, and edit their resumes.
- Then, students can view available internships and filter them based on various criteria.
- After selecting desired internships, students can initiate applications for internships and track the status of their applications (e.g., reviewed, shortlisted, selected).
- Furthermore, students can raise complaints or concerns regarding their internship experiences.

Company Features

- Companies must register to system with their credentials.
- Companies must log in to their account, and then, an create, edit, publish, and delete internship listings with all the necessary information (e.g. requirements, details of the job, ...)
- Then, for each internship posting, companies can browse and search through student profiles (CVs) that have applied for their internships.
- After receiving applications, companies can review incoming applications and shortlist candidates based on their qualifications.
- Then, companies can communicate with applicants and provide updates on application statuses.
- Companies can schedule and conduct interviews with shortlisted candidates and manage questionnaires for applicants.
- Finally, companies can decide which applicants are suitable for the positions based on interview and questionnaire.

Universities Features

- universities must register and login to their account.
- Then, universities can monitor the application processes of their students.
- Also, universities can oversee the execution of internships to ensure compliance and quality.

1.2.1 World Phenomena

coming soon!

1.2.2 Shared Phenomena

coming soon!

1.2.3 Machine Phenomena

coming soon!

1.3 Definitions, Acronyms, Abbreviations

1.3.1 Definitions

1.3.2 Acronyms

1.3.3 Abbreviations

1.4 Revision history

1.5 Reference Documents

1.6 Document Structure

2 Overall Description

2.1 Product perspective

2.1.1 Class Diagram

2.1.2 State Diagram

2.2 Product Functions

2.3 User Characteristics

2.3.1 Actors

2.4 Assumptions, Dependencies and Constraints

2.4.1 Domain Assumptions

2.4.2 Dependencies

2.4.3 Constraints

3 Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

3.1.2 Hardware Interfaces

3.1.3 Software Interfaces

3.1.4 Communication Interfaces

3.2 Functional Requirements

3.2.1 Use Case Diagram

3.2.2 Use Cases

3.2.3 Sequence Diagram

3.2.4 Goal Mapping on Requirements

3.3 Performance Requirements

3.4 Design Constraints

3.4.1 Standard Compliance

3.4.2 Hardware Limitations

3.4.3 Any Other Constraint

3.5 Software System Attributes

3.5.1 Reliability

3.5.2 Availability

3.5.3 Security